SMART SOLAR INVENTORY SYSTEM

Web-Based Inventory, Sales and Customer Service Platform
for Solar Equipment Shops

FINAL YEAR PROJECT PROPOSAL

Diploma of Associate Engineer (DAE) in Computer Information Technology
3rd Year

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1. Introduction

Solar energy is growing fast in Pakistan, and solar shops sell many products such as panels, inverters, batteries and charge controllers. Most of these shops still keep stock and sales records in paper registers, and customers have no easy way to see available products or contact the shop. The **Smart Solar Inventory System** solves both sides of this problem with one web-based platform made of two parts: a **Public Website** for customers and an **Admin Panel** for the shop owner and staff.

2. Problem Statement

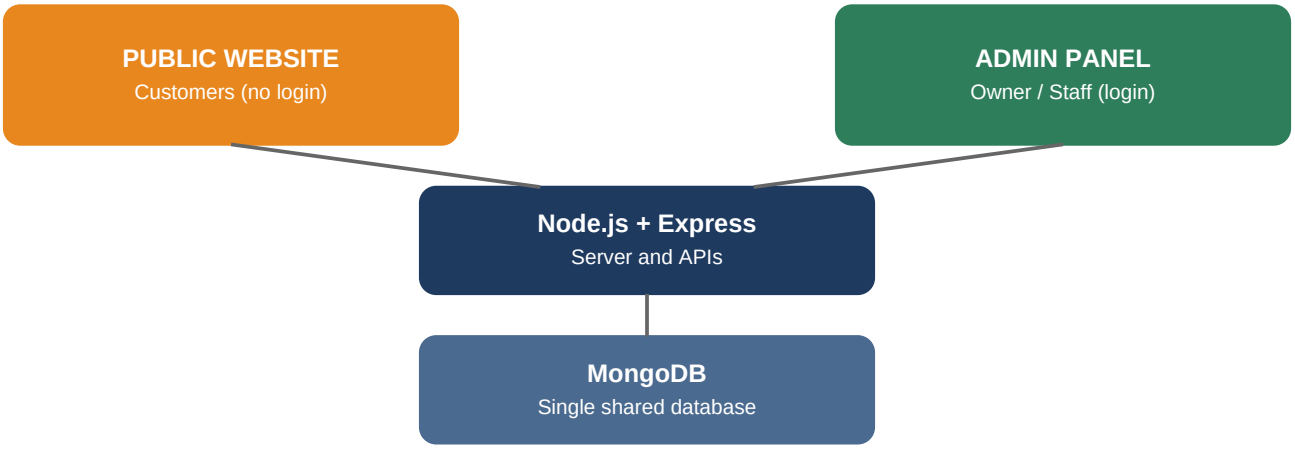
- Manual registers cause calculation mistakes and missing entries.
- The owner finds out an item is out of stock only when a customer asks for it.
- Serial numbers and warranty dates of inverters and batteries are not recorded properly.
- Customers cannot see products, prices or availability without visiting or calling the shop.
- Preparing daily or monthly sales and profit reports takes a lot of time.

3. Objectives

- To provide a public website where customers can view products, send order requests and check warranty.
- To provide a secure admin panel to manage products, stock, purchases, sales and orders.
- To update stock automatically after every purchase and sale.
- To alert the owner when an item falls below its minimum stock level.
- To track serial numbers and warranty expiry of sold products.
- To generate sales, purchase and profit reports through a dashboard, with print and export options.
- To add smart features: a Solar Calculator for customers and reorder suggestions for the owner.

4. System Overview

The system has two parts that share one database. Whatever the admin updates (stock, price) is shown instantly on the public website, and every customer order arrives in the admin panel.



5. Features

A. Public Website (for customers, no login required)

Page / Feature	Description
Home	Shop name, banner, featured solar products and offers.
Products	Catalog with product images, search, category filter, price and In Stock / Out of Stock status.
Solar Calculator	Customer enters appliances with watts and hours of use and gets a suggested system size (panels, inverter, battery).
Order Request	Customer enters name, phone, address and selected products; the order is sent to the admin panel.
Warranty Check	Customer enters a serial number to see warranty status and expiry date, and can submit a warranty claim.
About Us	Information about the shop and its services.
Contact and Helpline	Phone number, WhatsApp button, address and a query/complaint form.

B. Admin Panel (for Owner and Staff, login required)

Module	Description
Login and Roles	Secure login. Admin has full access; Staff can create bills and view stock only.
Dashboard	Today's sales, low stock items, pending orders and charts at a glance.
Product Management	Add, edit, delete products with category, brand, capacity (watt), prices and minimum stock level.
Stock Tracking	Live quantity of every product, updated automatically on purchase and sale.
Low Stock Alert	Highlights items below the minimum level.
Purchase and Suppliers	Record supplier details and purchase entries.
Order Management	View customer orders and change status: Pending, Confirmed, Delivered, Cancelled.
Sales and Invoice	Create customer bills and print or save invoices.
Warranty Tracking	Store serial numbers and warranty dates; list warranties ending soon.
Messages	View queries, complaints and warranty claims sent from the website.
Reports	Daily and monthly sales, profit and top-selling products.
Print and Export	Print invoices and export reports as PDF or Excel (CSV).
Activity Log	Records who added, changed or deleted what, for accountability.

6. Smart Features

These features make the system more than a simple record keeper. Each one uses simple logic that can be built with JavaScript and MongoDB queries.

Smart Feature	How it works
Solar Calculator (customers)	Customer lists appliances (fans, lights, fridge, AC) with watts and hours of use. The system adds up the load and suggests a suitable inverter size, number of panels and battery capacity, then shows matching products from the shop's stock.
Reorder Suggestion (owner)	The system looks at the average sales of the last 3 months for each product and suggests how many units to buy again before stock runs out.
Low Stock Alert	When quantity falls below the minimum level, the item is highlighted on the dashboard.
Warranty Reminder	Shows products whose warranty ends within the next 30 days so the owner can inform customers.
Top and Slow Products	Dashboard charts show best-selling and slow-moving products to guide purchasing decisions.

The Solar Calculator gives an estimate only; final system design should be confirmed by the shop's technician.

7. How an Order Works

Step	What happens
1	Customer opens the website and browses available products.
2	Customer submits an order request with name, phone and address.
3	The order appears in the admin panel as **Pending** and the owner calls the customer to confirm.
4	Owner marks the order **Confirmed**, creates the bill and stock decreases automatically.
5	After delivery the order is marked **Delivered** and the warranty record is saved.

Payment is Cash on Delivery or paid at the shop. Online payment gateways are kept outside the project scope.

8. Technologies Used

Layer	Technology
Front-end	HTML, CSS, JavaScript, Bootstrap (responsive design for mobile and desktop)
Back-end	Node.js with Express.js
Database	MongoDB
Tools	VS Code, MongoDB Compass, Postman, Git
Hosting (optional)	Demo runs on a laptop (localhost); can be put online using free tiers such as Render (server) and MongoDB Atlas (database)

All tools are free and open source, so the project needs no financial investment.

9. System Requirements

Type	Requirement
Software	Windows/Linux, Node.js, MongoDB (or MongoDB Atlas), VS Code, a modern browser (Chrome / Edge)
Hardware (development)	Laptop/PC with 4 GB RAM or more, Intel Core i3 or above, 5 GB free disk space
Users	Shop owner/staff and customers with any mobile or computer and an internet browser

10. Database Design (MongoDB Collections)

Collection	Main Fields
users	name, email, password (hashed), role
products	name, category, brand, watt, purchasePrice, salePrice, quantity, minStock
suppliers	name, phone, address
purchases	supplier, items, quantity, cost, date
orders	customerName, phone, address, items, status, date
sales	customer, items (with serial numbers), quantity, total, date
warranties	product, serialNumber, saleDate, expiryDate
messages	name, phone, message, date
activityLogs	user, action, item, date

11. Security and Testing

- **Security:** passwords stored in hashed form (bcrypt), login with tokens (JWT), role-based access, and input validation on all forms.
- **Testing:** each module is tested for correct input and wrong input; APIs are tested with Postman; stock calculation is checked with sample sales and purchases; the website is checked on mobile and desktop.

12. Scope and Limitations

- Designed for a single shop; multi-branch support is future work.
- Orders are confirmed by phone and paid on delivery or at the shop; no online payment gateway.
- The Solar Calculator provides an estimate and does not replace a technician's site survey.
- Reorder suggestions depend on past sales records; sample data is used for the demonstration.

13. Project Timeline

Phase	Work	Duration
1	Requirement gathering, proposal, system design and database design	2 weeks
2	Admin panel: login, products, stock, purchases, sales	4 weeks
3	Public website: home, products, order request, contact	3 weeks
4	Orders, alerts, warranty tracking, reports, print/export, activity log	3 weeks
5	Smart features: Solar Calculator and reorder suggestion	2 weeks
6	Testing, bug fixing, optional free hosting and documentation	2 weeks
7	Final presentation and report	1 week

Total: about 17 weeks. The core system (Phases 2 to 4) is completed and tested first; smart features are added once the core is stable.

14. Expected Outcomes

- A working website where customers can view products, use the Solar Calculator, send orders and check warranty.
- An admin panel that replaces paper-based stock registers.
- Fewer stock-out situations through low stock alerts.
- Quick sales and profit reports for better business decisions.
- Organized warranty records that improve customer service.

15. Future Scope

- Online payment (JazzCash / Easypaisa / card) and delivery tracking.
- Customer credit (udhaar) and payment ledger for regular customers.
- Barcode / QR code scanning for fast product entry and billing.
- WhatsApp or SMS alerts for orders, low stock and warranty reminders.
- Demand prediction from previous sales data.
- Multi-branch support and a mobile application.

16. Conclusion

The Smart Solar Inventory System gives solar shops a complete digital solution: a public website that brings customers closer to the shop, and an admin panel that keeps stock, sales and warranty records accurate. It uses free, widely used web technologies and is practical to complete within the final year timeline.